

Andrew Chen

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Work Experience

SOFTWARE ENGINEER - UIC PAYMENTS INC.

OCT 2024 - JULY 2026

- Sole developer for the Bezel8-SP unattended payment terminal, developing and maintaining embedded C applications on Linux for payment processing, device configuration, networking, diagnostics, and cloud terminal management.
- Optimized QA feature-testing time by developing on-device diagnostic tools, accelerating provisioning across hundreds of devices.
- Reduced TLS handshake time from ~10 seconds to 2–3 seconds by adding ECDSA support and optimizing Mbed-TLS configurations, cutting boot time by up to 50%.
- Developed bespoke payment applications, including an unattended car-wash flow that managed machine-control pulse signals, timed re-crediting for uninterrupted sessions, configurable settings, and end-to-end payment processing.
- Diagnosed and fixed embedded payment crashes caused by edge cases in transaction flows, memory handling bugs and startup failures, preventing over \$5,000 in RMA shipping and repair costs.
- Designed and implemented a Voyager fleet payment processing core within the payment engine, routing required EMV L2 and L3 data, handling ISO 8583 messages, and adding a custom transaction flow to capture driver identification data in compliance with official Voyager specifications.
- Created Interac Canada debit transaction handlers for purchase and refund flows, implementing specification-mandated data fields and debit-specific state handling within the existing payment engine.
- Led the complete certification efforts for Voyager fleet cards and Interac debit by coordinating with certification advisors, executing test cases, resolving kernel issues, and preparing required documentation, achieving full production certification for both payment cores.
- Built network management and configuration flows for Linux-based terminals, supporting DHCP/static IP, DNS, gateway, and subnet settings across multiple hardware generations.

Projects

EYEMATIC - AUTOMATIC IRIS DETECTION

AUG 2023 - JUNE 2024

- Customized PolarFire SoC FPGA firmware to support real-time neural network detection of iris, pupil, and surgical instruments with >95% confidence for cataract surgery assistance.
- Trained Tiny-YOLOv3 on 1,000+ annotated frames of surgical footage and generated custom FPGA firmware to support the full inference pipeline, including camera input and HDMI passthrough.

Education

UNIVERSITY OF CALIFORNIA, SANTA BARBARA

SEP 2020 - JUNE 2024

- Bachelor of Science in Computer Engineering

Technical Skills

- **Languages:** C, C++, Python, Java
- **Software Development:** Embedded Linux, Payment Terminals, Network Management, API Integration, Diagnostic Tooling
- **Tools & Methodologies:** Git, Docker, REST APIs, Cross-Compilation, EMV, ISO 8583